



Our **EXPERIENCE**
For your **DIGITAL**
TRANSFORMATION



Agenda

Big Data and Context

Lambda architectures

MapReduce to Spark

Data Structures

Streaming Caveats

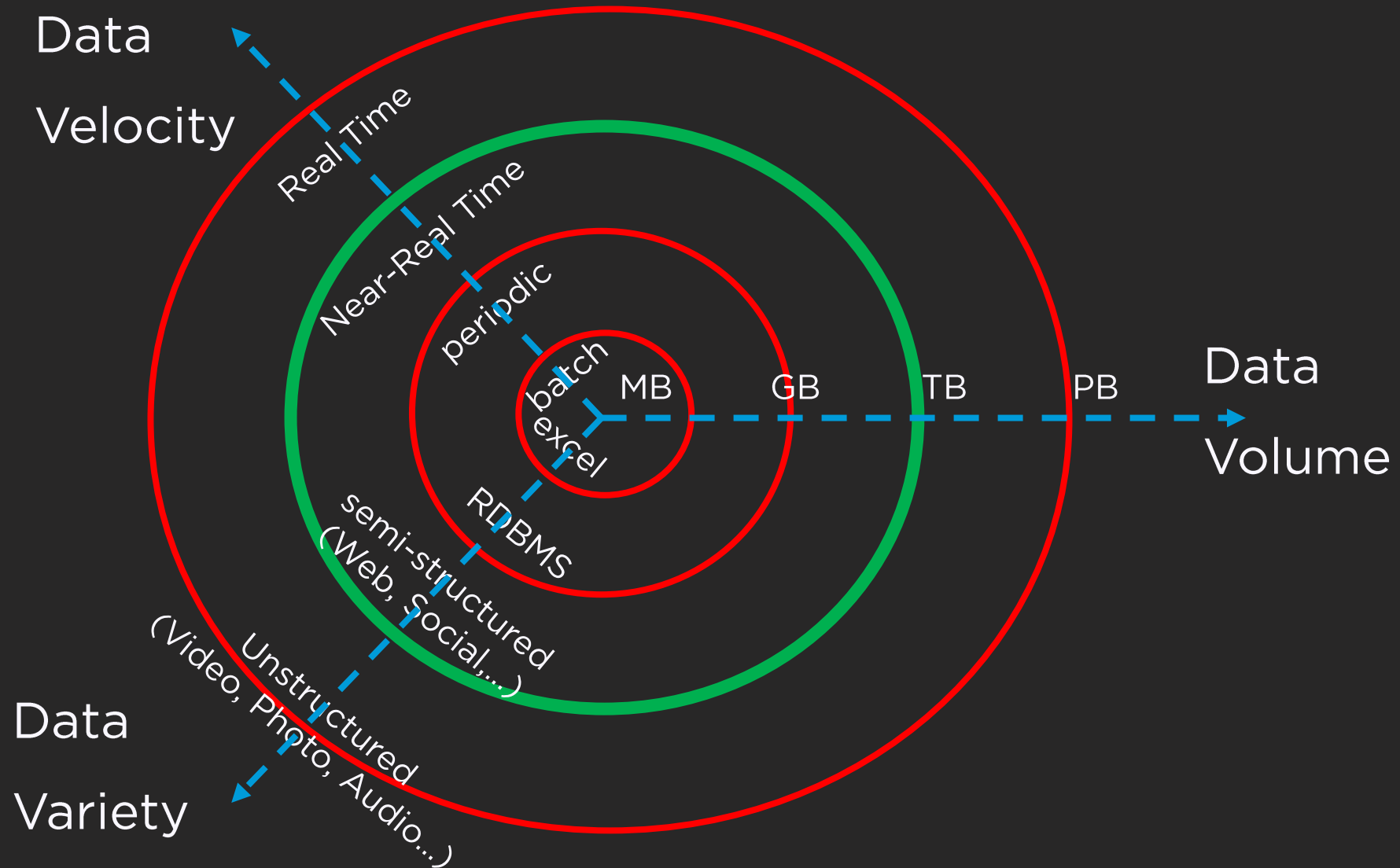
What's next?

Do you really have BIG DATA?

«Big data is like teenage sex;
Everyone talks about it,
Nobody really knows how to do it,
Everyone thinks everyone else is doing it,
So everyone claims they are doing it».

Dan Ariely, Duke University

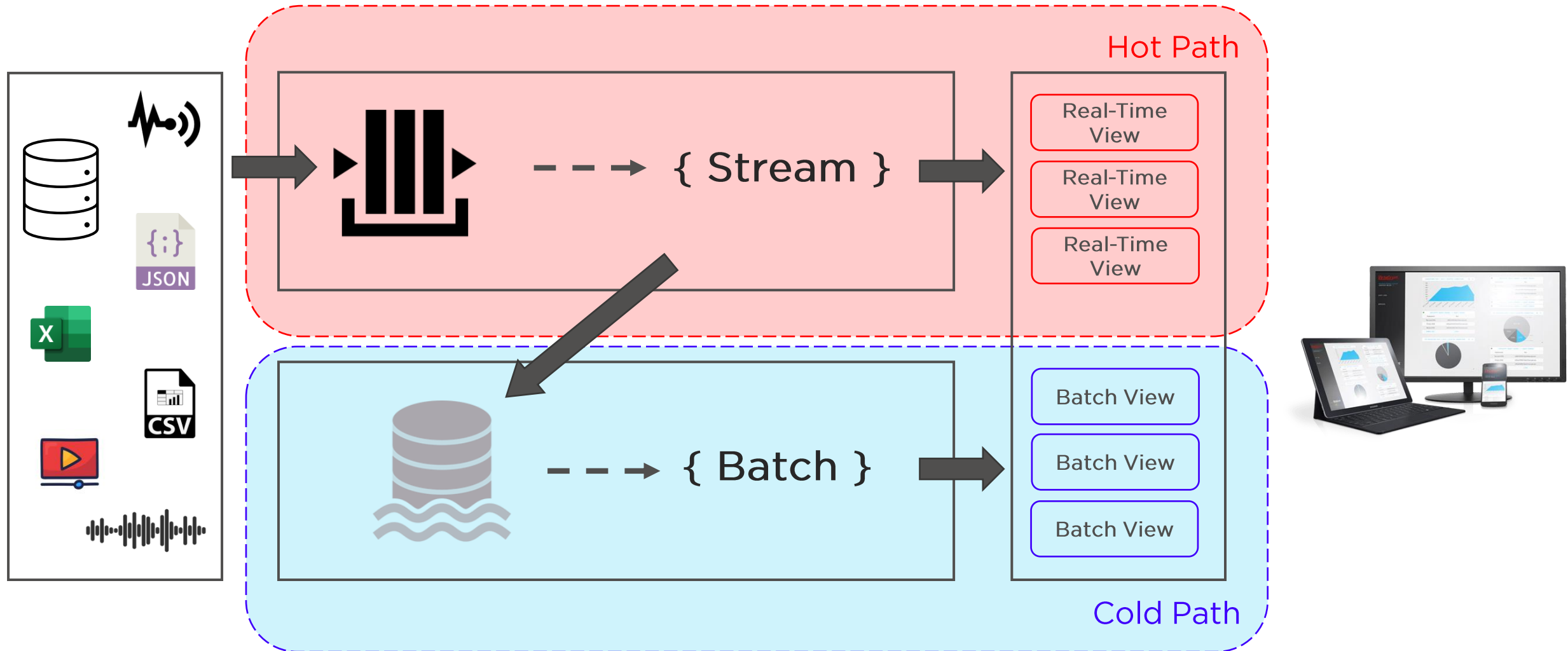
Big Data THE 3Vs CONCEPT



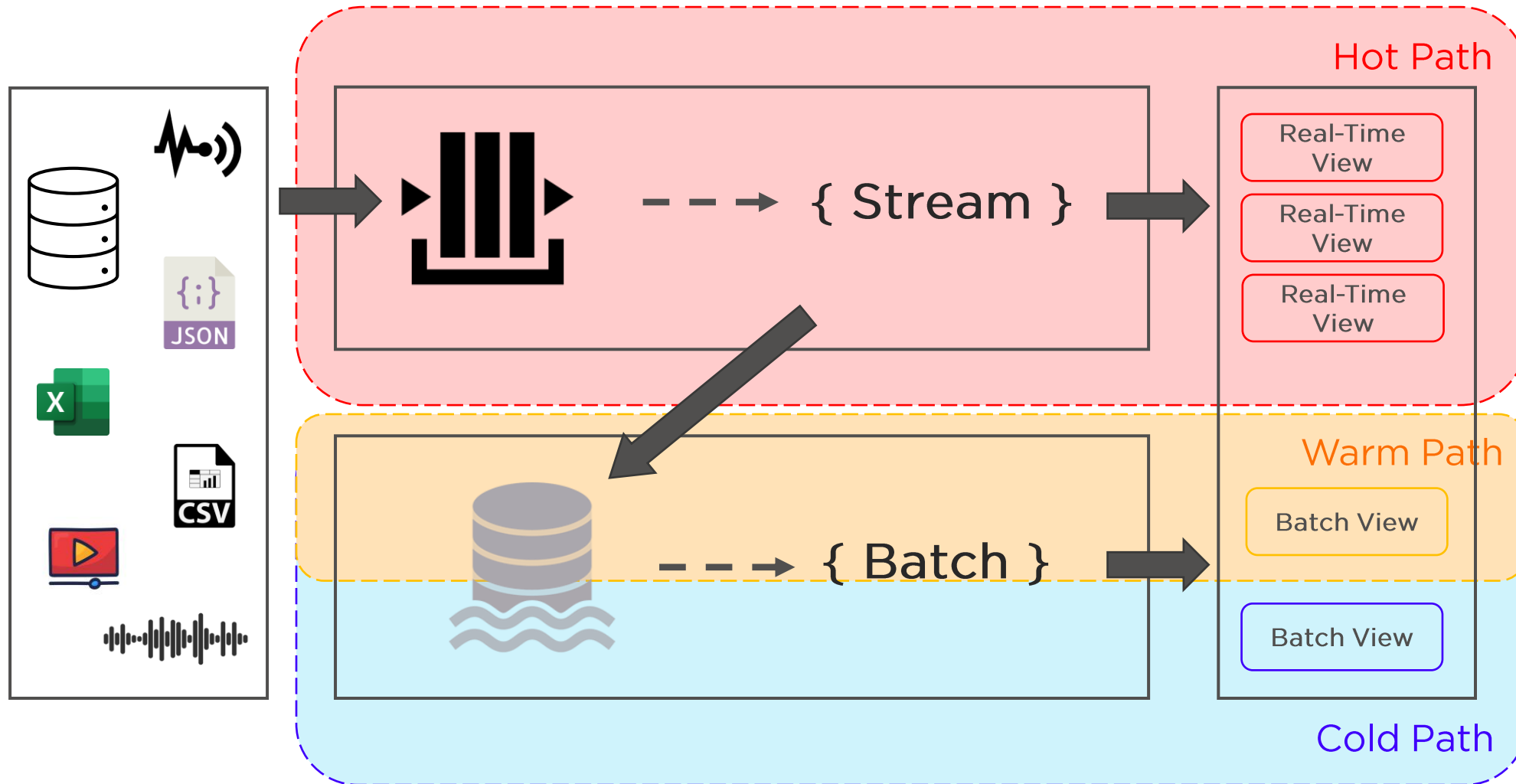
Context



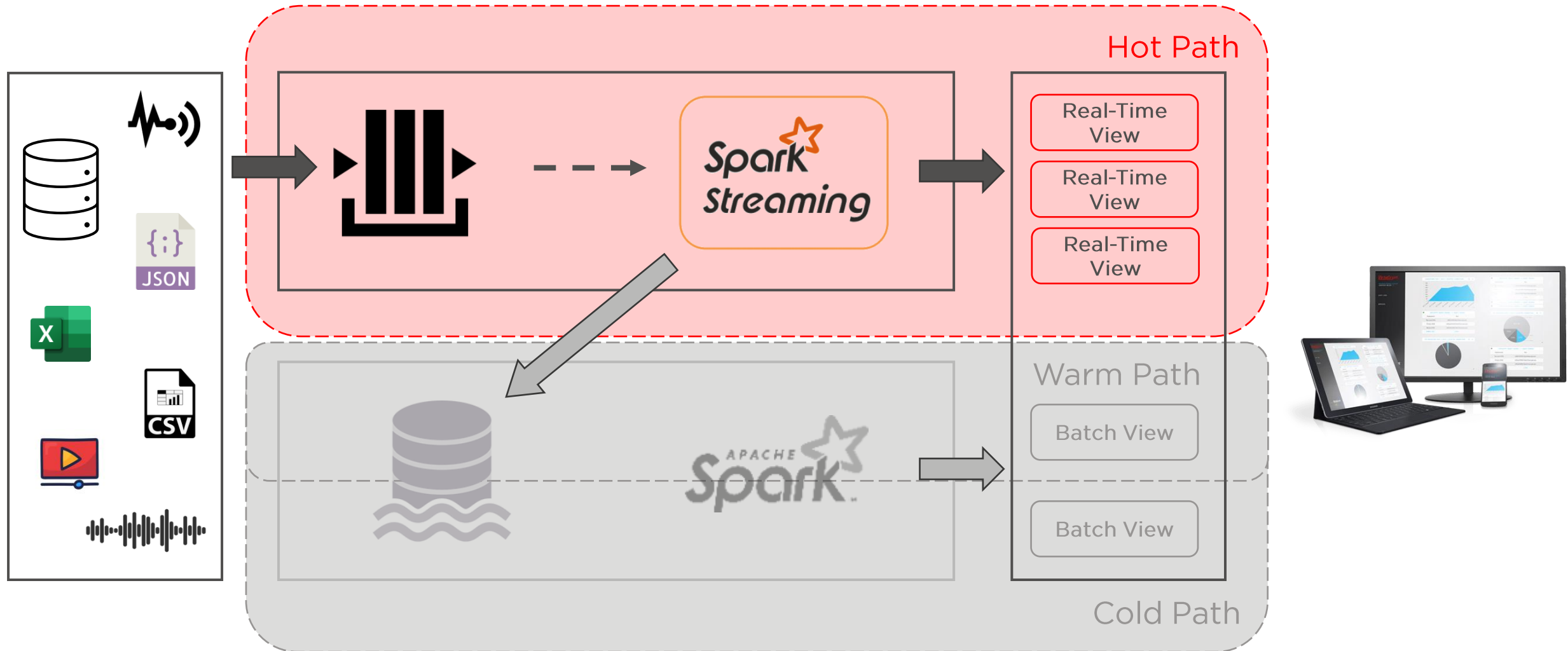
Lambda ARCHITECTURE



3-ways Lambda ARCHITECTURE

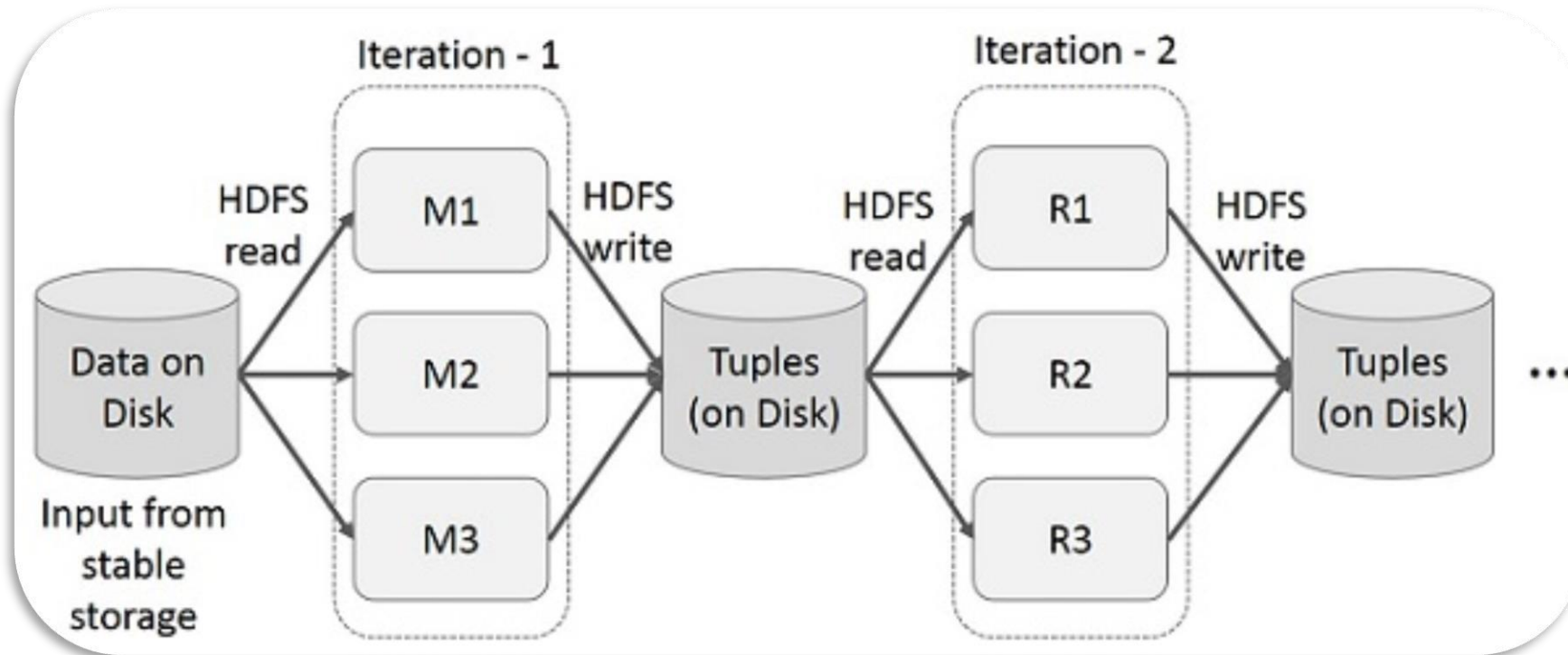


Hot PATH



MapReduce

It is a parallel processing system for managing huge amounts of data, following the *divide et impera* strategy.



MapReduce paradigm

For each batch:

1. Read from storage
2. Map
3. Write to storage

Drawbacks of this approach

- too many I/O operations
- not ideal for streaming

Streaming

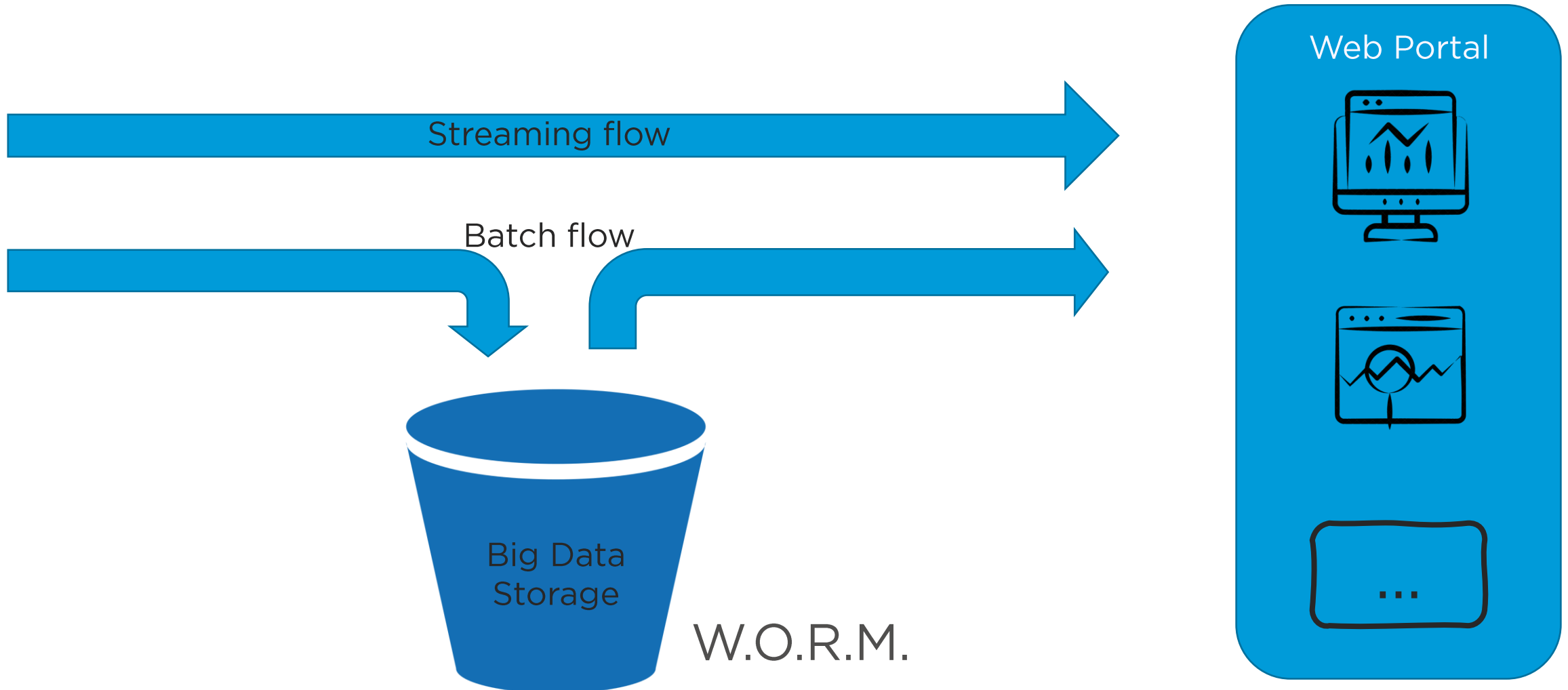
Kafka Streaming

- Provides low latency (ms)
- Best for kafka-to-kafka applications
- Advanced fault tolerance
- Hyper-scalable

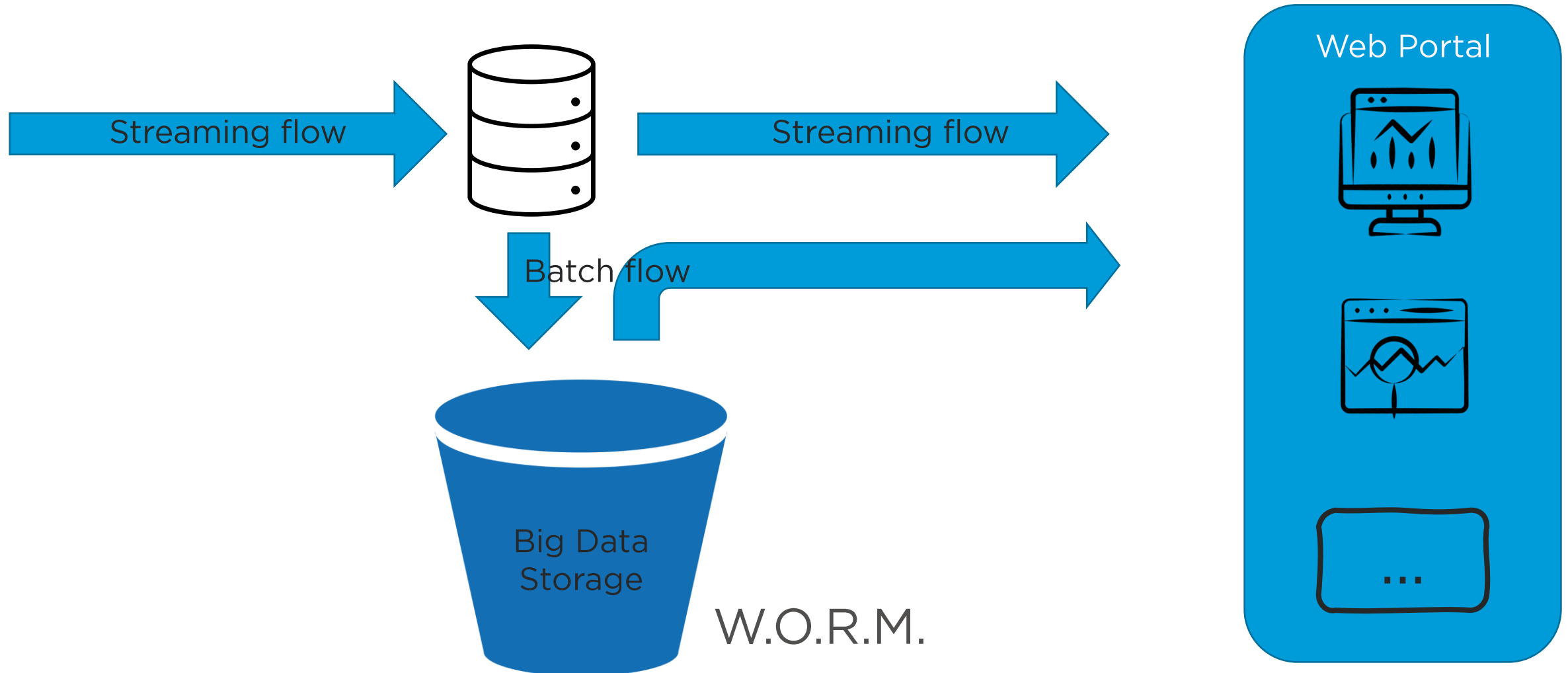
Spark Streaming

- Higher latency (seconds)
- Best solution for different systems integration
- Fault tolerant too
- Scalable too but requires more resources
- Code flexibility (same code for streaming and batch)

Storage types CASE HISTORY



Storage types CASE HISTORY



Spark

It is an open-source distributed general-purpose **cluster-computing** framework, like MapReduce.

In-memory computation

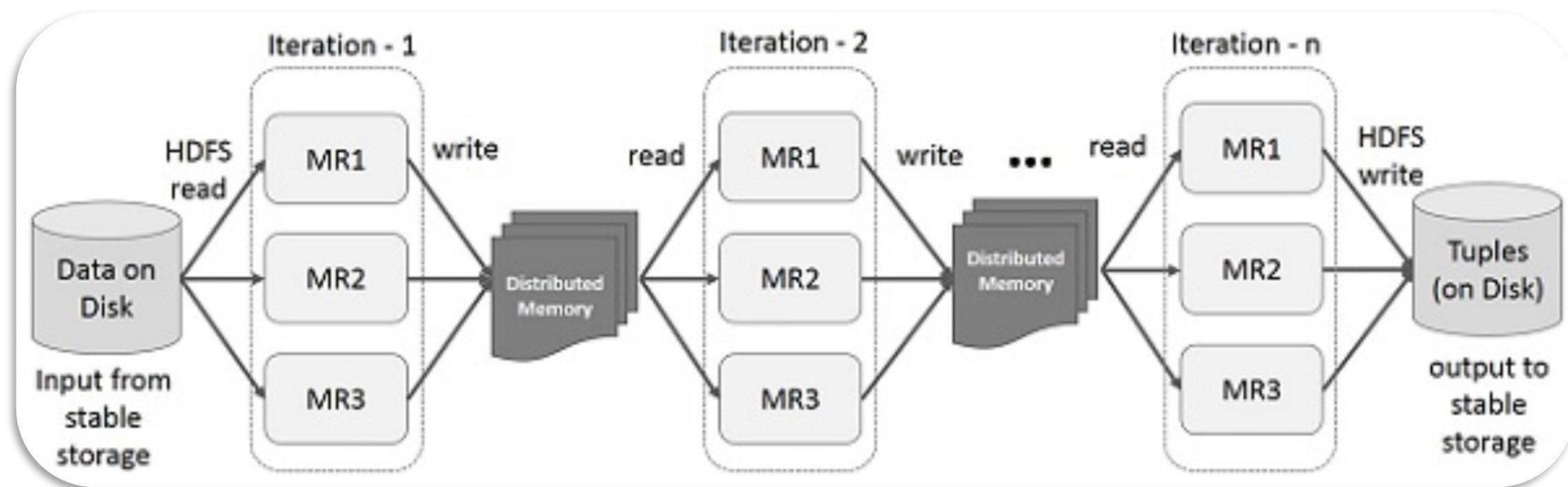
As a result, it is capable of running programs up to 100x faster than MapReduce, making it suitable for real-time computation.

Lazy evaluation

It means that the execution of transformations isn't triggered until there is an action that needs results. Therefore, the sequence of operations are analyzed for optimization.

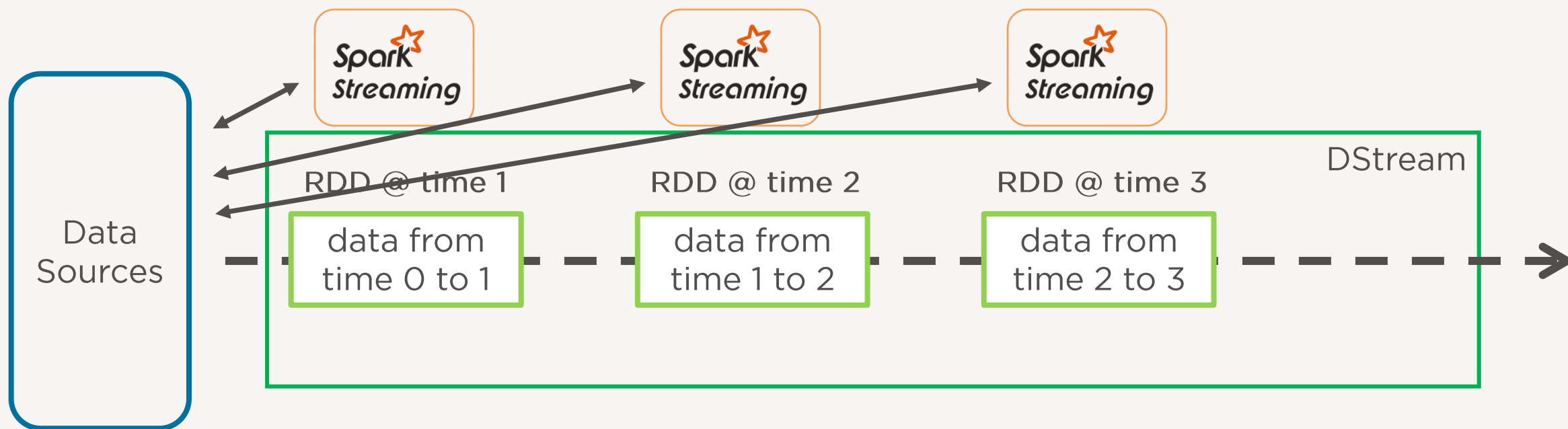
RDDs

RDDs evolution



Spark STREAMING

DStreams



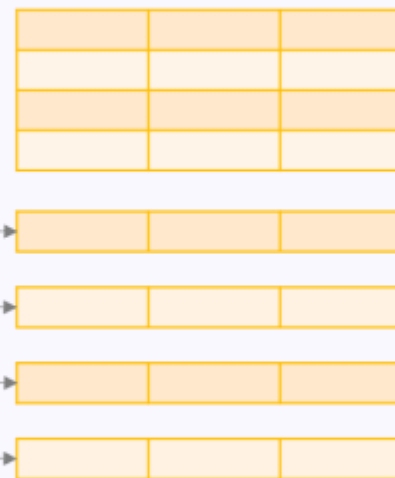
Dataframes and Structured STREAMING

	Col 1	Col 2	Col 3	Col 4
Row 1				
Row 2				
Row 3				
Row 4				
Row 5				
Row 6				

Data stream



Unbounded Table

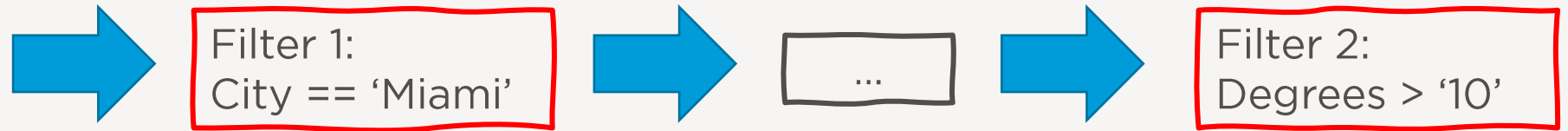


new data in the
data stream
=
new rows appended
to a unbounded table

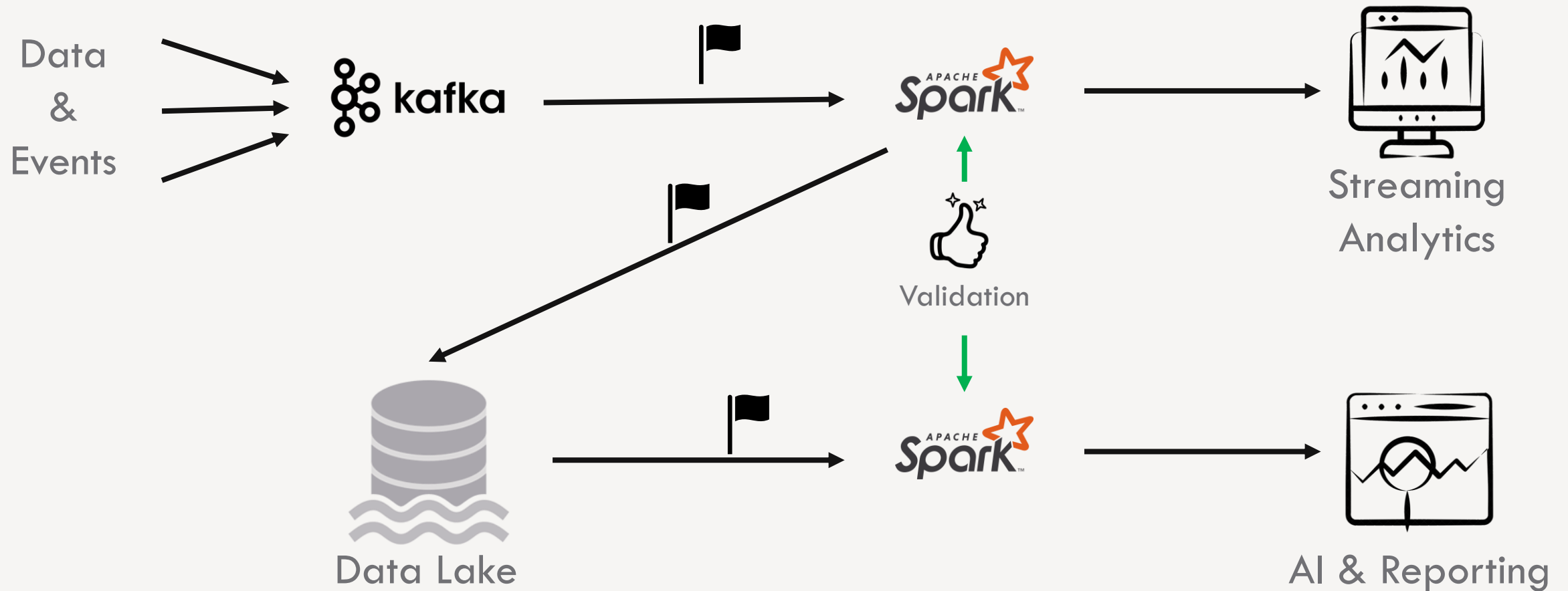
Data stream as an unbounded table

RDDs execution plan

```
RDD  
[  
  ('New York', 10),  
  ('Chicago', 15),  
  ('Miami', 8),  
  ('New York', 2),  
  ('Chicago', 7),  
  ('Miami', 19)  
]
```

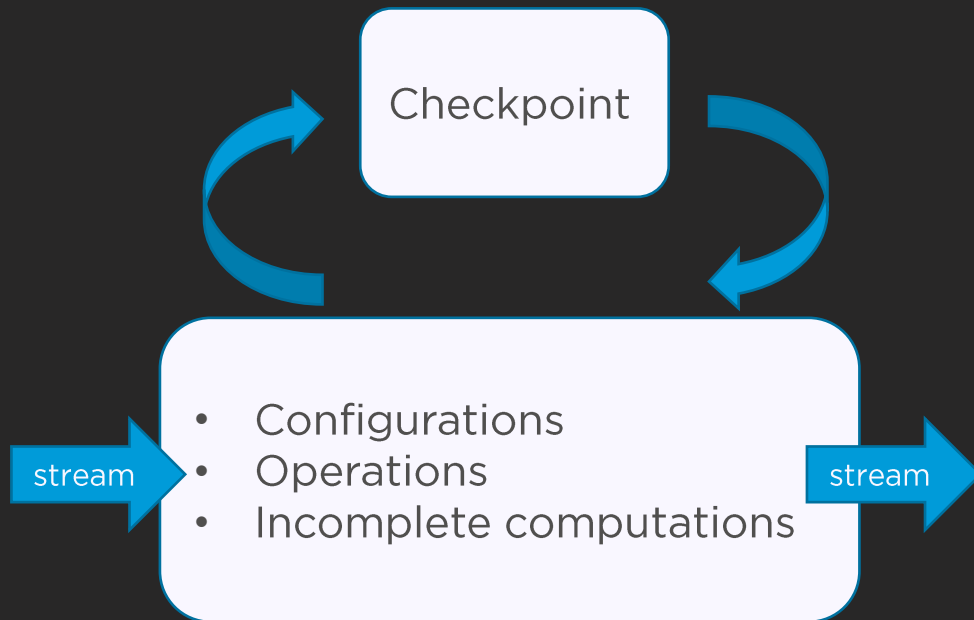


Flow CONTROL

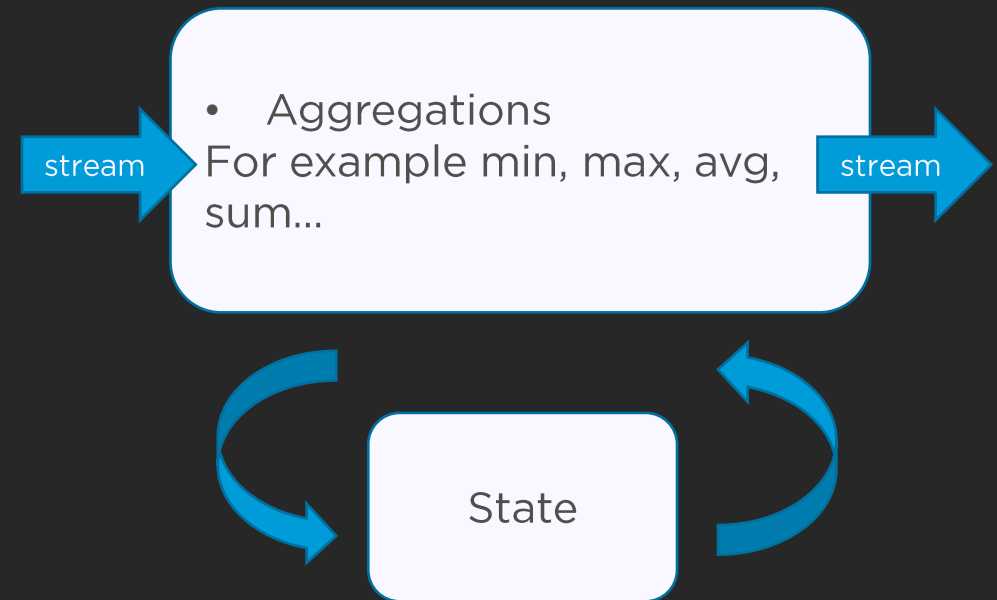


Checkpoints & State

Checkpoint



State



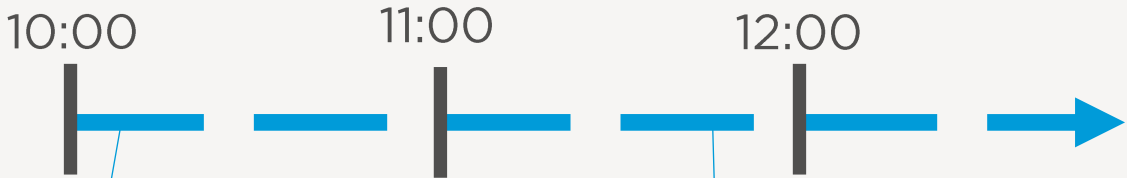
Windows

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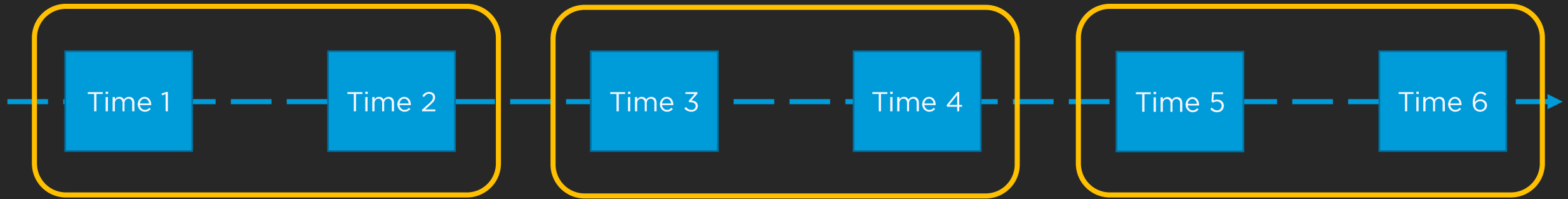


10:02	Piece 1	Steel temp	1140
10:02	Piece 1	Torque	22.6
10:04	Piece 1	Steel temp	1152
10:04	Piece 1	Torque	22.5
...	...	...	...

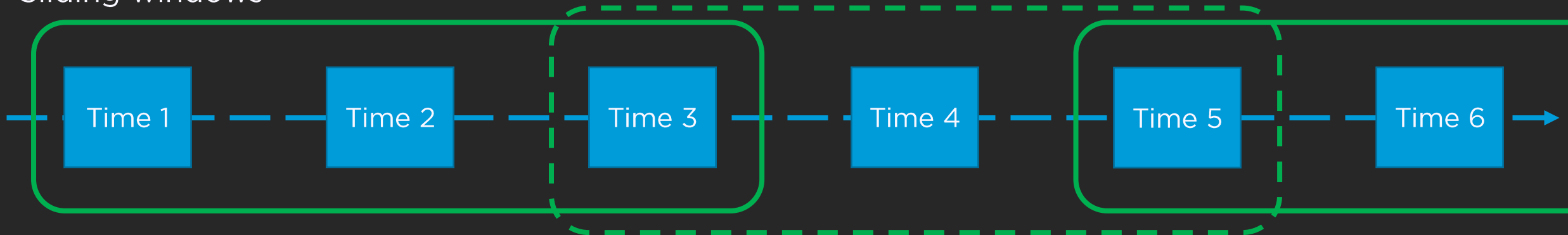
11:46	Piece 2	Steel temp	1137
11:46	Piece 2	Torque	22.2
11:48	Piece 2	Steel temp	1163
11:48	Piece 2	Torque	22.9
...	...	...	...

Window TYPES

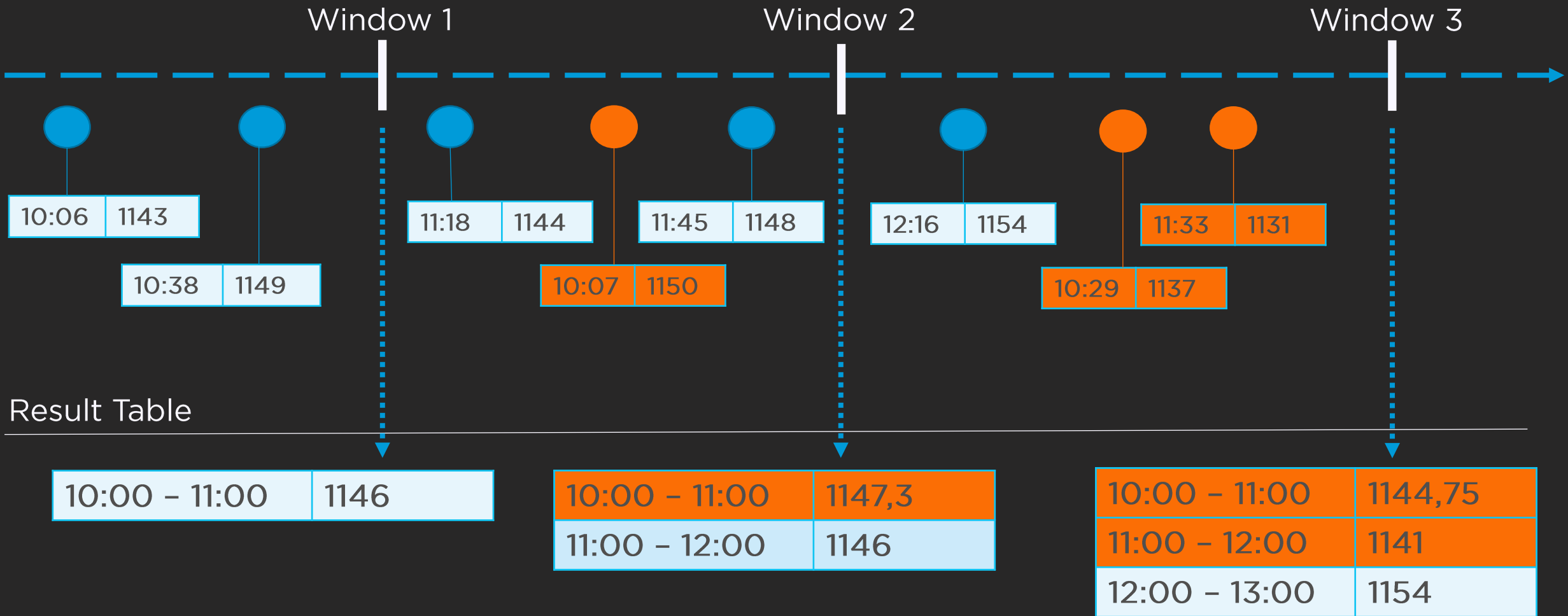
Tumbling windows



Sliding windows

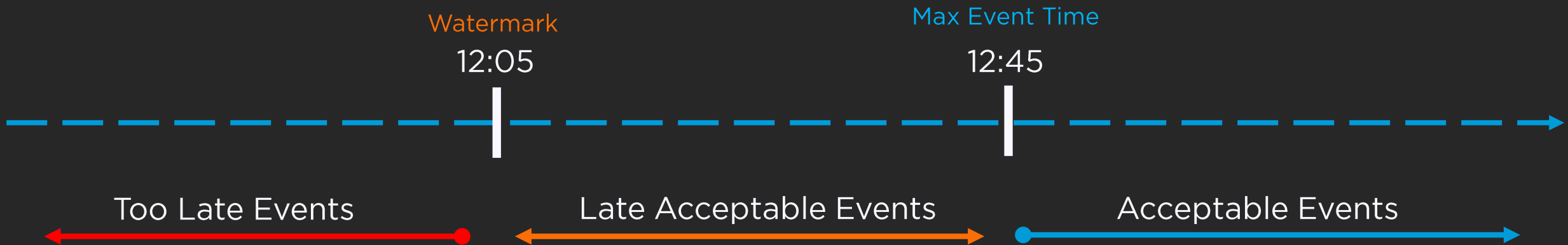


Late DATA



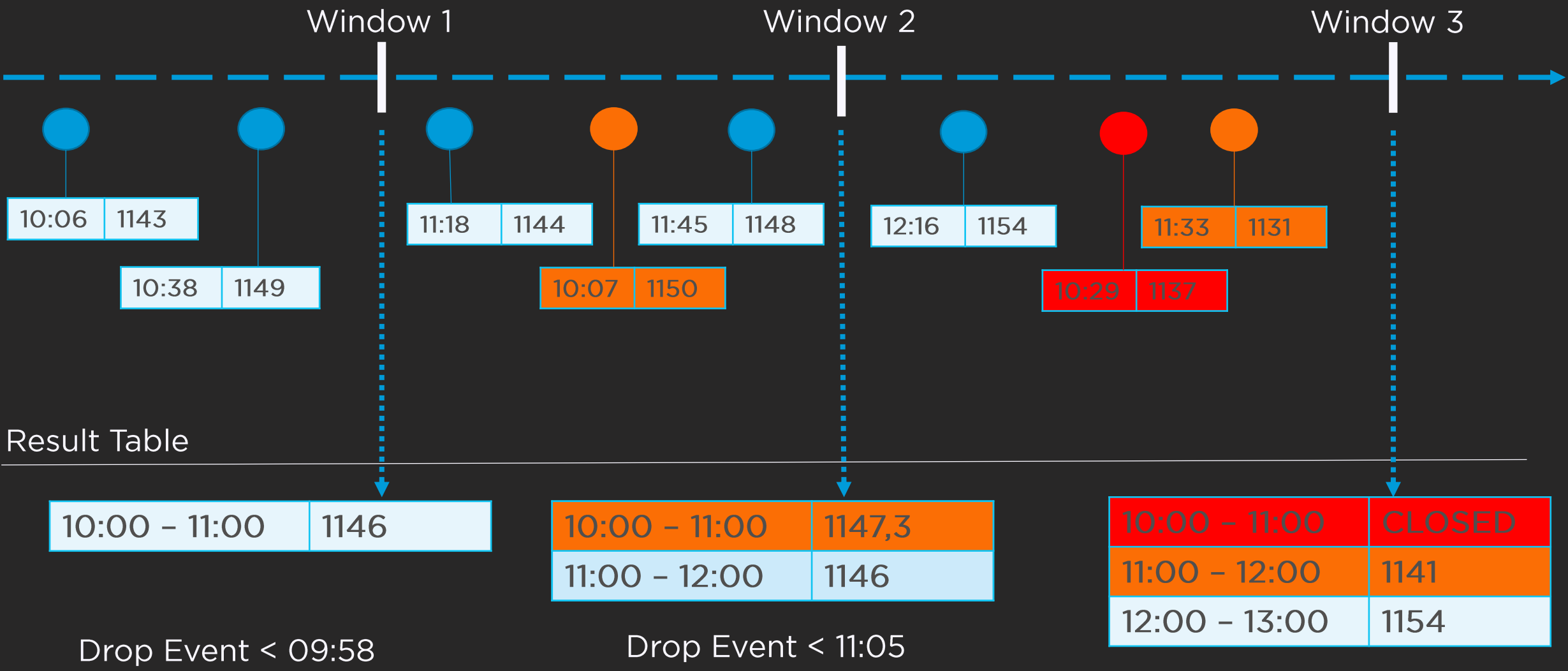
Watermarks

Watermark value = 40 mins



Watermarks

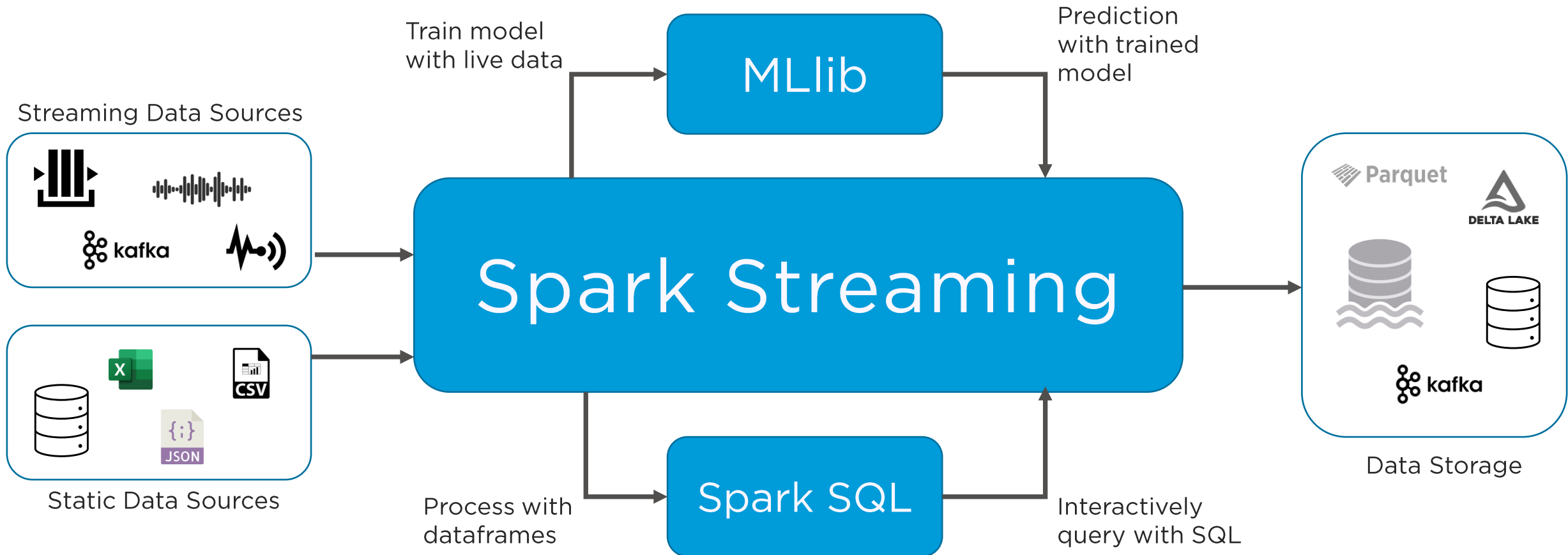
Watermark value = 40 mins



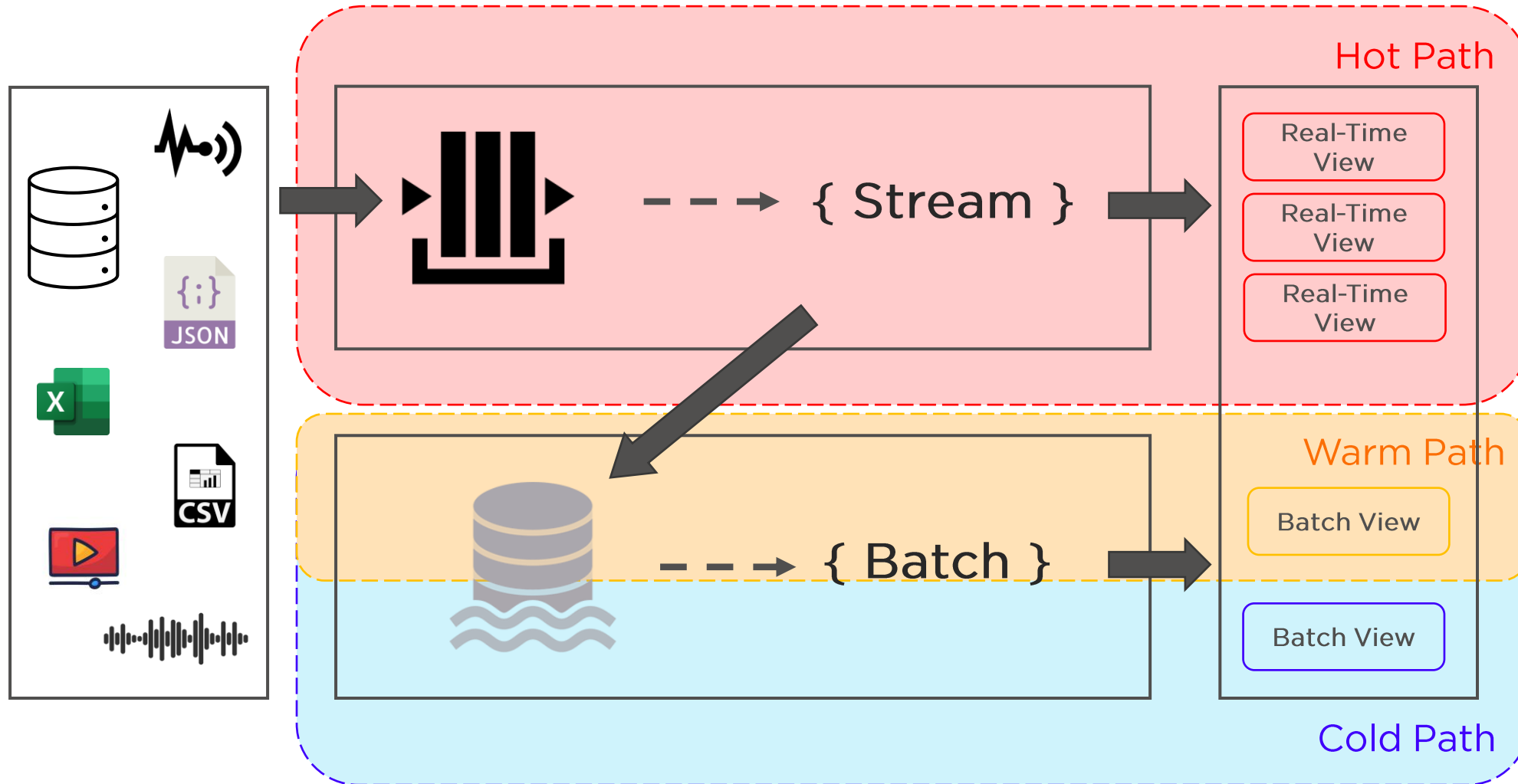


DEMO

Spark STRUCTURED STREAMING



Close the CIRCLE



BUSINESS ANALYTICS PER IL **MANUFACTURING**



DANIELI AUTOMATION

COLLABORATORS

419

2019 REVENUES

195 ML

COUNTRIES

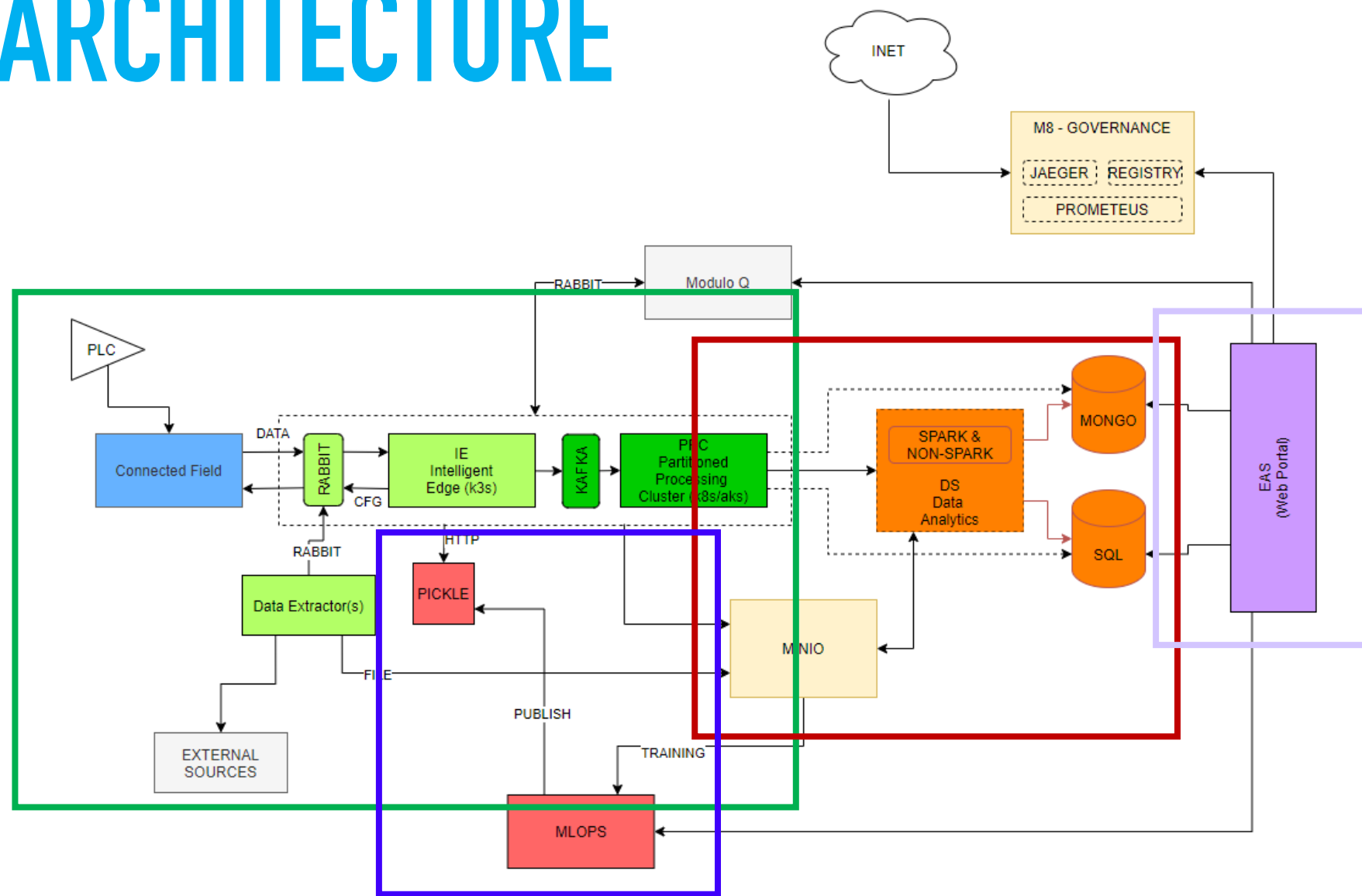
Italia - Slovenia - Croazia - Tailandia - Cina - USA
Europa - Canada - Russia - Vietnam - India - Taiwan
Giappone

CASE NEEDS

Develop a vertical platform for metal industry built to ingest, store and analyze both process data and business data.

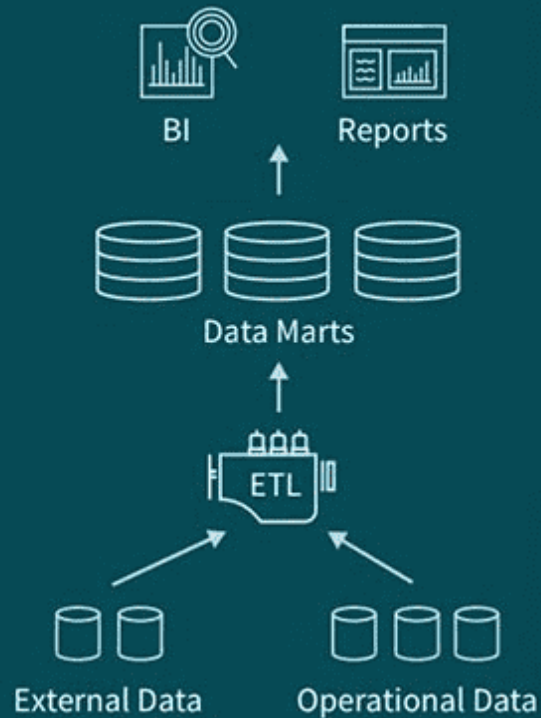
The main focus had to be on the production cycle starting from order acquisition to delivery through all the transformation phases of the raw material to final product. The solution we provided is at the forefront and embraces the Measure - Gain Insights - Improve philosophy..

Danieli ARCHITECTURE

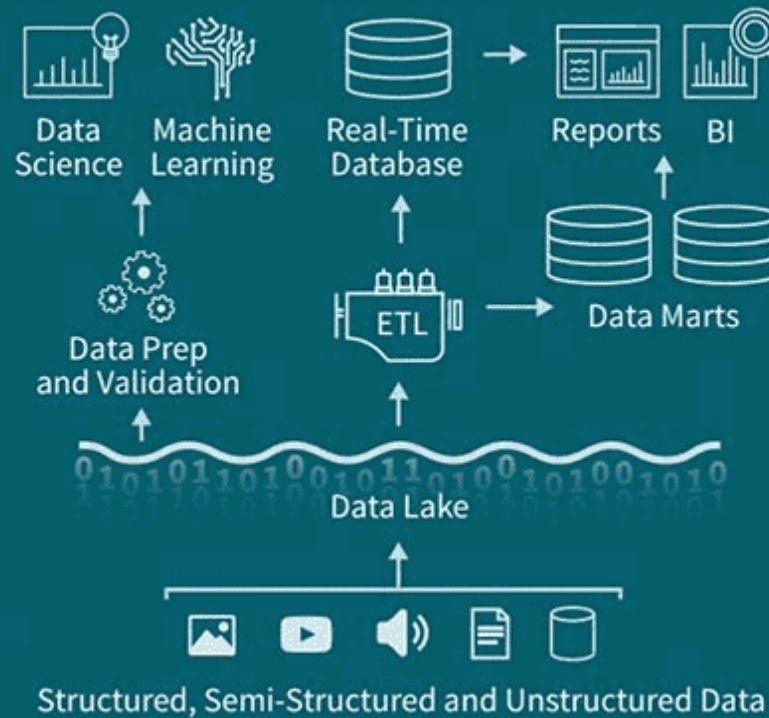


What's NEXT?

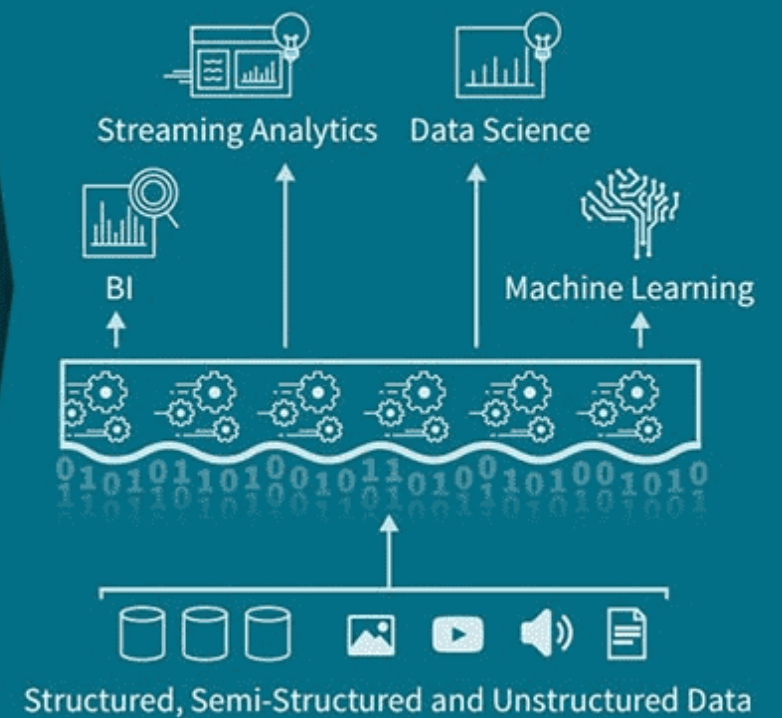
Data Warehouse



Data Lake

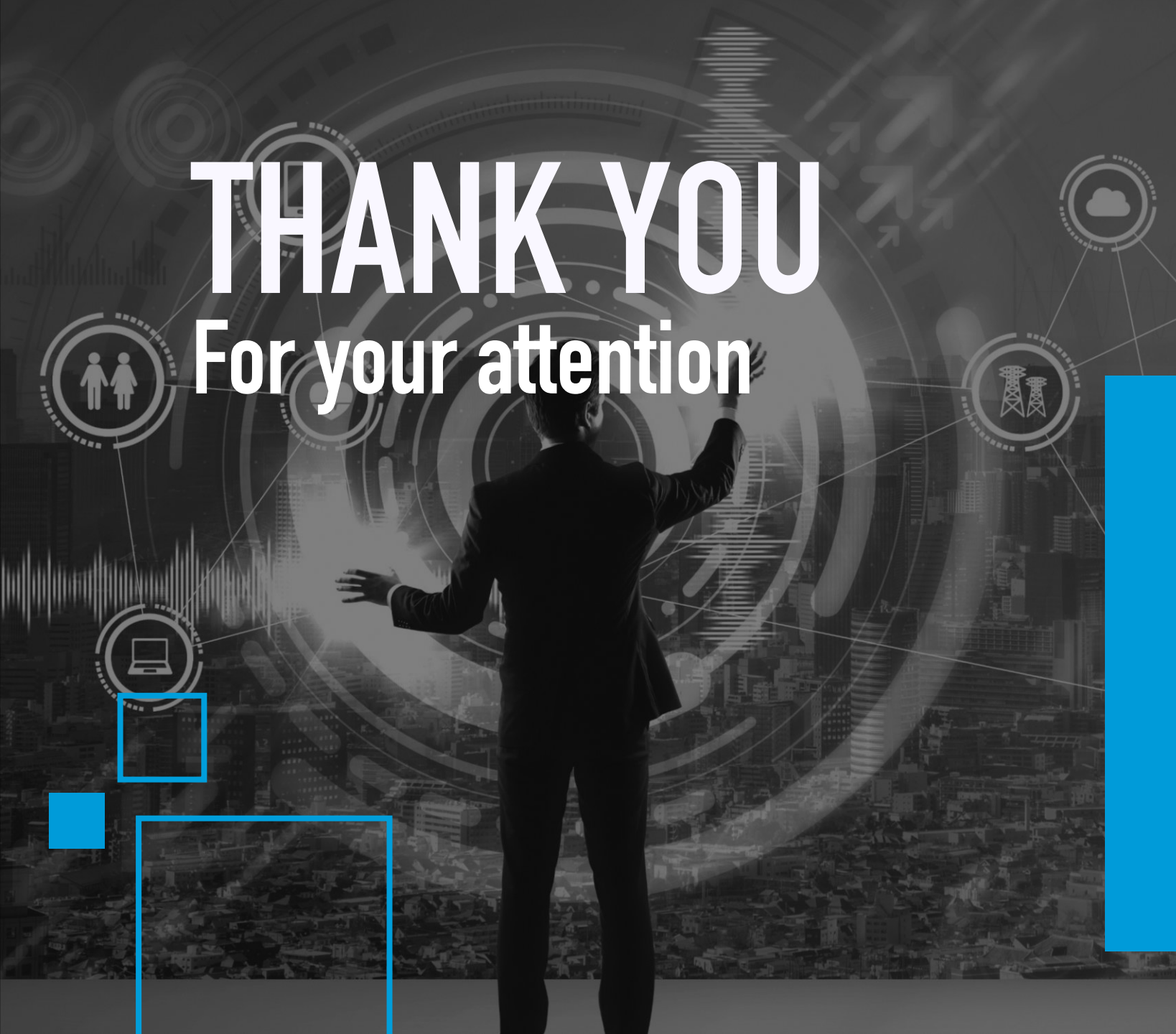


Lakehouse



THANK YOU

For your attention



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